

Draft Solutions and Marking Guide

Ph.D. Macroeconomics II, Questions 2–4

1 Question 1: Optimal Subsidies in the Quality Ladders Model

Take the quality ladders model as studied in lectures, with some modifications. The physical marginal cost of making a machine of variety ν at quality q is given by $\psi q(\nu, t | q)$. However, the government pays a subsidy $S_M \in (0, 1)$ such that the actual marginal cost faced by machine producers is $\psi_e q(\nu, t | q)$, where $\psi_e \equiv (1 - S_M)\psi$.

The government also pays a subsidy $S_{R\&D} \in (0, 1)$ for R&D expenditure. (Assume that the government budget is balanced by imposing lump-sum taxes on consumers.) Let us also set $\theta = 1$.

1. The final-good production function is $Y(t) = \frac{1}{1-\beta} L^\beta \int_0^1 q(\nu, t) x(\nu, t)^{1-\beta} d\nu$. Show that the demand curve for a given variety is $x(\nu, t | q) = \left(\frac{q(\nu, t)}{p(\nu, t | q)} \right)^{1/\beta} L$.

2. The owner of the patent for variety ν at quality q maximizes profits $\pi(\nu, t | q) = p(\nu, t | q) x(\nu, t | q) - \psi_e q(\nu, t | q) x(\nu, t | q)$. Imposing the normalization $1 - \beta \equiv \psi$ from now on, show that $p(\nu, t | q) = (1 - S_M)q(\nu, t)$ and $\pi(\nu, t | q) = (1 - S_M)^{\frac{\beta-1}{\beta}} \beta q(\nu, t) L$.

3. Let $V(\nu, t | q)$ be the value of a patent that a successful innovator obtains if she raises the quality of variety ν to the level $q(\nu, t)$. Spending one unit of the final good on research generates a flow rate of innovation η/q^I , where $q^I = \frac{q(\nu, t)}{\lambda}$ is the incumbent's quality level. Interpret the condition for free entry into research, $1 - S_{R\&D} = \frac{\eta V(\nu, t | q)}{q(\nu, t)/\lambda}$.

4. Interpret the no-arbitrage condition $r(t)V(\nu, t | q) = \pi(\nu, t | q) - z(\nu, t | q)V(\nu, t | q) + \dot{V}(\nu, t | q)$

$q)$.

5. Explain why under free entry, $\dot{V}(\nu, t \mid q) = 0$. Use the free entry and the no-arbitrage conditions, and the expression for profits you derived to show that $r(t) + z(\nu, t \mid q) = s\lambda\eta\beta L$, where $s \equiv \frac{(1-S_M)^{\frac{\beta-1}{\beta}}}{1-S_{R\&D}}$.

6. On a balanced growth path (BGP), the following equations must hold: $r + z = s\lambda\eta\beta L$, $g = r - \rho$, and $g = (\lambda - 1)z$. Show that $g = \frac{s\lambda\eta\beta L - \rho}{(\lambda-1)^{-1}+1}$.

7. The socially optimal growth rate is $g^* = (\lambda - 1)(1 - \beta)^{-1/\beta}\eta\beta L - \rho$. Show that to achieve this growth rate, it must be that $s = (1 - \beta)^{-1/\beta} - \frac{\rho}{(\lambda-1)\lambda\eta\beta L}$.

8. True or false? Explain.

“Any combination of S_M and $S_{R\&D}$ that satisfies equation $s = (1 - \beta)^{-1/\beta} - \frac{\rho}{(\lambda-1)\lambda\eta\beta L}$ will achieve the socially optimal allocation.”

Question 2: Government spending in the New-Keynesian model

1. Goods clearing implies

$$\hat{c}_t = \frac{\hat{y}_t - s_G \hat{g}_t}{1 - s_G}.$$

Using $\hat{n}_t = \hat{y}_t$ and $\widehat{mc}_t = \hat{c}_t + \varphi \hat{n}_t$,

$$\widehat{mc}_t = \left(\frac{1}{1 - s_G} + \varphi \right) \hat{y}_t - \frac{s_G}{1 - s_G} \hat{g}_t.$$

Flexible prices imply $\widehat{mc}_t = 0$, hence

$$\hat{y}_t^n = \frac{s_G}{1 + \varphi(1 - s_G)} \hat{g}_t.$$

Give up to 2 points for goods clearing/private consumption, 2 points for marginal cost, and 1 point for natural output.

2. Let

$$D \equiv 1 + \varphi(1 - s_G), \quad \psi \equiv \frac{s_G}{D}.$$

Then $\hat{y}_t^n = \psi \hat{g}_t$. The natural level of private consumption is

$$\hat{c}_t^n = \frac{\hat{y}_t^n - s_G \hat{g}_t}{1 - s_G} = -\frac{\varphi s_G}{D} \hat{g}_t.$$

The natural real interest rate is

$$r_t^n = E_t \hat{c}_{t+1}^n - \hat{c}_t^n = (1 - \rho_g) \frac{\varphi s_G}{D} \hat{g}_t.$$

The output gap is $x_t = \hat{y}_t - \hat{y}_t^n$. Since the private-consumption gap is $x_t/(1 - s_G)$, the gap IS equation is

$$x_t = E_t x_{t+1} - (1 - s_G)(\hat{y}_t - E_t \pi_{t+1} - r_t^n).$$

The Phillips curve is

$$\pi_t = \beta E_t \pi_{t+1} + \kappa_y x_t, \quad \kappa_y \equiv \kappa \left(\frac{1}{1 - s_G} + \varphi \right) = \kappa \frac{D}{1 - s_G}.$$

Give 2 points for defining the gap and the Phillips curve, 2 points for the correctly scaled IS equation, and 1 point for r_t^n .

3. Guess $x_t = a_x \hat{g}_t$ and $\pi_t = a_\pi \hat{g}_t$. Since $E_t \hat{g}_{t+1} = \rho_g \hat{g}_t$,

$$(1 - \rho_g) a_x + (1 - s_G)(\phi_\pi - \rho_g) a_\pi = (1 - s_G)(1 - \rho_g) \frac{\varphi s_G}{D},$$

and

$$(1 - \beta \rho_g) a_\pi = \kappa_y a_x.$$

Thus

$$a_x = \frac{(1 - s_G)(1 - \rho_g) \frac{\varphi s_G}{D}}{(1 - \rho_g) + (1 - s_G)(\phi_\pi - \rho_g) \frac{\kappa_y}{1 - \beta \rho_g}},$$

and

$$a_\pi = \frac{\kappa_y}{1 - \beta \rho_g} a_x.$$

Both coefficients are positive under the stated assumptions. A higher ϕ_π raises the denominator and therefore lowers a_x and a_π . Give 3 points for the algebraic solution and 2 points for signs/comparative statics.

4. Under flexible prices, $x_t = 0$, $\pi_t = 0$, and output rises by

$$\hat{y}_t^n = \frac{s_G}{D} \hat{g}_t.$$

Private consumption falls because government purchases absorb resources and create a negative wealth effect:

$$\hat{c}_t^n = -\frac{\varphi^{SG}}{D} \hat{g}_t < 0.$$

Under sticky prices and the Taylor rule, the government-spending shock raises the natural real rate. Since the nominal-rate rule does not perfectly replicate the natural real rate, the real rate is too low relative to r_t^n , so $x_t > 0$ and $\pi_t > 0$. Output rises by more than under flexible prices:

$$\hat{y}_t = \hat{y}_t^n + x_t.$$

Private consumption may fall by less than under flexible prices, and can in principle rise if the positive output gap is large enough, since

$$\hat{c}_t = \hat{c}_t^n + \frac{x_t}{1 - s_G}.$$

Optimal monetary policy stabilizes the output gap and inflation, implementing $x_t = 0$ and $\pi_t = 0$ by setting the real rate equal to the natural real rate. Give 2 points for flexible prices, 2 points for sticky-price mechanism, and 1 point for optimal policy.

Question 3: Recruiting intensity in the DMP model

1.

$$\begin{aligned} rW &= w + \sigma(U - W), & rU &= b + \lambda_u(\Theta)(W - U), \\ rJ &= y - w + \sigma(V - J), & rV &= -c - \frac{\kappa e^{1+\psi}}{1+\psi} + e\lambda_v(\Theta)(J - V), \end{aligned}$$

where the last equation holds at the firm's optimal choice of e . Give 1 point for the worker equations, 1 point for the job-value equation, and 1 point for the vacancy-value equation with recruiting intensity.

2. A stationary symmetric equilibrium is a tuple

$$\{W, U, J, V, w, e, \Theta, u, \mathcal{V}\}$$

such that the Bellman equations hold; e solves the firm's recruiting-intensity problem; $V = 0$ (free entry); wages satisfy Nash bargaining,

$$\gamma(J - V) = (1 - \gamma)(W - U);$$

aggregate recruiting tightness is consistent,

$$\Theta = \frac{e\mathcal{V}}{u};$$

and unemployment satisfies the Beveridge relation

$$u = \frac{\sigma}{\sigma + \lambda_u(\Theta)}.$$

Give 1 point for optimization/free entry/wage setting, 1 point for aggregate consistency, and 1 point for the steady-state unemployment relation.

3. The interior recruiting-intensity condition is

$$\kappa e^\psi = \lambda_v(\Theta)(J - V).$$

Free entry implies $V = 0$, so

$$\kappa e^\psi = \lambda_v(\Theta)J, \quad c + \frac{\kappa e^{1+\psi}}{1+\psi} = e\lambda_v(\Theta)J, \quad c = \frac{\psi}{1+\psi} \kappa e^{1+\psi}.$$

The job-creation curve is

$$\frac{y - w}{r + \sigma} = \frac{\kappa e^\psi}{\lambda_v(\Theta)}.$$

Relative to the standard DMP model, job creation depends on both market tightness and recruiting intensity through the term $\kappa e^\psi / \lambda_v(\Theta)$. Give 1 point for the recruiting-intensity FOC, 1 point for free entry and the cost relation for e , and 1 point for the modified job-creation curve and interpretation.

4. With $V = 0$, worker and firm surpluses are

$$W - U = \frac{w - b}{r + \sigma + \lambda_u(\Theta)}, \quad J = \frac{y - w}{r + \sigma}.$$

Nash bargaining implies

$$w = (1 - \gamma)b + \gamma[y + \lambda_u(\Theta)J] = (1 - \gamma)b + \gamma[y + \Theta \kappa e^\psi].$$

Higher recruiting intensity raises wages by improving the worker's outside option through a higher job-finding rate. Give 2 points for the wage equation and 1 point for the interpretation.

5. A lower κ raises recruiting intensity,

$$e = \left(\frac{(1 + \psi)c}{\psi\kappa} \right)^{1/(1+\psi)},$$

lowers the effective recruiting cost κe^ψ , and therefore raises Θ , lowers u , and raises w . Intuition: cheaper recruiting makes search more intensive, the labor market becomes tighter, unemployment falls, and workers capture part of the gain through bargaining. Give 1 point for e , 1 point for Θ and u , and 1 point for w and intuition.

Question 4: Idiosyncratic risk in the Aiyagari model

1. The household problem is

$$V(a, \epsilon) = \max_{c, a'} \{u(c) + \beta E[V(a', \epsilon')|\epsilon]\}$$

subject to

$$c + a' = w\epsilon + (1 + r)a, \quad a' \geq 0, \quad c \geq 0.$$

For an interior choice of assets, the Euler equation is

$$u'(c(a, \epsilon)) = \beta(1 + r)E[u'(c(a', \epsilon'))|\epsilon],$$

with the usual inequality version when the borrowing constraint binds.

The firm solves

$$\max_{K, L} K^\alpha L^{1-\alpha} - (r + \delta)K - wL.$$

The first-order conditions are

$$r + \delta = \alpha K^{\alpha-1} L^{1-\alpha}, \quad w = (1 - \alpha) K^\alpha L^{-\alpha}.$$

With mean labor efficiency equal to one, $L = 1$ in stationary equilibrium. Give 1 point for the household problem, 1 point for the household Euler condition, 1 point for the firm problem, and 1 point for the firm FOCs.

2. A stationary competitive equilibrium consists of prices (r, w) , household decision rules $c(a, \epsilon)$ and $a'(a, \epsilon)$, firm choices (K, L) , and an invariant distribution $\mu(a, \epsilon)$ such that: households

solve their problem; firms optimize; μ is invariant under the household policy and the Markov process for ϵ ; labor clears with $L = \int \epsilon d\mu = 1$; asset/capital markets clear with

$$K = \int a d\mu;$$

and the goods market clears,

$$\int c(a, \epsilon) d\mu + \delta K = K^\alpha L^{1-\alpha}.$$

Give up to 4 points for these equilibrium components.

3. The Aiyagari diagram plots aggregate asset supply $A^s(r)$ from household saving behavior and capital demand $K^d(r)$ from firm optimality. Capital demand is downward sloping:

$$K^d(r) = \left(\frac{\alpha}{r + \delta} \right)^{1/(1-\alpha)}$$

when $L = 1$. Aggregate asset supply is increasing in r : a higher return makes saving more attractive. With uninsured idiosyncratic risk and prudence, households have a precautionary-saving motive. As r approaches $1/\beta - 1$ from below, desired saving becomes very large. A stationary equilibrium therefore requires

$$r < 1/\beta - 1.$$

Give 1 point for demand, 1 point for supply/diagram, and 1 point for the interest-rate bound.

4. More idiosyncratic income risk strengthens precautionary saving. For any given r , households want to hold more assets, so $A^s(r)$ shifts to the right. In equilibrium, the interest rate falls and the capital stock rises. The higher capital stock raises wages and lowers the marginal product of capital. In a representative-agent or complete-markets economy with the same mean labor endowment, purely idiosyncratic risk is insured or diversified away, so this precautionary-saving channel is absent. Give 2 points for the shift and equilibrium effects, and 2 points for the complete-markets contrast.